Maesuring Power of Star-Free Languages

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Abstract

A language L is said to be $\mathcal{C}$-measurable, where $\mathcal{C}$ is a class of languages, if there is an infinite sequence of languages in $\mathcal{C}$ that "converges" to L. In this paper, we investigate the measuring powers of SF of the class of all star-free languages. It is shown that the measuring power of star-free languages are coincides with the measuring power of generalised definite languages (a more restricted subclass of star-free languages), which implies the decidability of SF-measurability for regular languages. We also show that every regular language can be represented as a convergent sequence of Boolean combinations of star-free and group languages, *i.e.*, every regular language is $\mathscr{B}(\mathsf{SF} \cup \mathsf{G})$-measurable where G is the class of all group languages and $\mathscr{B}\mathcal{C}$ is the Boolean algebra generated by $\mathcal{C}$. We also demonstrate several consequences of the above two important facts.

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1 Introduction

Approximating a complex object by more simple objects is a major concept in both computer science and mathematics. In the theory of formal languages, various types of approximations have been investigated (*e.g.*, [10, 11, 6, 4, 3, 5]). For example, Kappes and Kintala [10] introduced *convergent-reliability* and *slender-reliability* which measure how a given deterministic automaton $\mathcal{A}$ nicely approximates a given language L over an alphabet A. A slightly modified version of approximation notion is *bounded-ϵ-approximation* which was introduced by Eisman and Ravikumar [7]. They say that two languages L_1 and L_2 provide a bounded-ϵ-approximation of language L if $L_1 \subseteq L \subseteq L_2$ holds and the ratio of their length-n difference satisfies $\#((L_2 \setminus L_1) \cap A^n) / \#(A^n) \leq \epsilon$ for every sufficiently large $n \in \mathbb{N}$. Also, there is a different framework of approximation so-called *minimal-cover* [5, 3], and a notion represents some *in*approximability by regular languages so-called REG-*immunity* [8]. A model of approximation studied in this paper is $\mathcal{C}$-*measurability*, which was introduced in [14]. This notion is rather close to bounded-ϵ-approximation. Instead of approximating by a *single* regular language, we consider an approximation of some non-regular language L by an *infinite sequence* of regular languages that "converges" to L. Intuitively, we say that L is $\mathcal{C}$-*measurable* if there exists an infinite sequence of pairs of languages $(K_n, M_n)_{n\in\mathbb{N}}$ in $\mathcal{C}$ such that $K_n \subseteq L \subseteq M_n$ holds for all n and the "size" (*density*) of the difference $M_n \setminus K_n$ tends to 0 if n tends to infinity. The notion $\mathcal{C}$-measurability can be considered as a non-commutative and high-dimensional extension of Buck's measure theoretic approach to the density on natural numbers [2].

So far, the decidability and different characterisation of $\mathcal{C}$-measurability is systematically studied for some subclass $\mathcal{C}$ of regular languages. For example, two simple and decidable

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characterisations of PT-measurable and AT-measurable languages are given in [16], where PT is the class of all piecewise testable languages and AT is the class of all alphabet testable languages (*i.e.*, languages definable by the first-order logic with only one variable). For the class GD of generalised definite languages, an automata theoretic characterisation of the GD-measurability for regular languages is given in [17] and it is decidable in quadratic time [9] with respect to the size of given deterministic automaton. Some properties of the measuring power of star-free languages is investigated in [15], but the decidability for regular languages was left open. The decidability of G-measurability for regular languages is still unknown, while the measurability of two restricted subclasses Gcom (*commutative* group languages) and MOD (languages defined by length-modulo condition) of group languages are shown to be decidable for regular languages [19].

The main results of this paper are the following two theorems. The former one gives the decidable characterisation of SF-measurability for regular languages, and the latter one gives a measure theoretic "decomposition theorem" of regular languages into star-free and regular languages.

1. A language L is SF-measurable if and only if L is GD-measurable (Theorem 8).
2. Every regular language is $\mathscr{B}(\mathsf{SF} \cup \mathsf{G})$-measurable (Theorem 13).

2 Preliminaries

This section provides the precise definitions of density, measurability, considered subclasses of regular languages. REG_A denotes the family of all regular languages over an alphabet A. For a word $w \in A^*$ and a letter $a \in A$, we write $|w|_a$ the number of occurrences of a in w. The complement of L over A is denoted by $L^{\mathrm{c}} = A^* \setminus L$. We assume that the reader has a standard knowledge of algebraic language theory (*cf.* [12, 13]).

The main targets of this paper are the following three subclasses of regular languages. The *star-free languages*, the *generalised definite languages* GD and the *group languages* G:

$$\mathsf{SF}_A \stackrel{\text{def}}{=} \{\, L \subseteq A^* \mid L \text{ is recognised by a finite aperiodic monoid} \,\}$$

$$\mathsf{GD}_A \stackrel{\text{def}}{=} \mathscr{B}\{wA^*, A^*w \mid w \in A^*\}$$

$$\mathsf{G}_A \stackrel{\text{def}}{=} \{\, L \subseteq A^* \mid L \text{ is recognised by a finite group} \,\}$$

A monoid M divides a monoid N if M is a monoid homomorphic image of a submonoid of N. It is well-known that a monoid M recognises a language L if and only if the syntactic monoid M_L of L divides M (*cf.* [12, Theorem 10.2.6]), hence L is in SF_A (G_A, respectively) if and only if M_L is a finite aperiodic monoid (a finite group, respectively). It is easy to see that a language L is generalised definite if and only if $L = E \cup \bigcup_{(u,v)\in F} uA^*v$ for some finite sets E and $F \subseteq A^* \times A^*$. Let FIN_A be the class of all finite and co-finite languages over A. Hereafter, we only consider an alphabet A such that $\#(A) \geq 2$ (because $\mathsf{SF}_A = \mathsf{GD}_A = \mathsf{FIN}_A$ holds if $\#(A) = 1$) and sometimes omit the subscript A for denoting these classes of languages.

2.1 Density and measurability of formal language

For a set X, we denote by $\#(X)$ the cardinality of X. We denote by $\mathbb{N}$ the set of natural numbers including 0.

▶ **Definition 1** (*cf.* [1])**.** The *density* $\delta_A(L)$ of $L \subseteq A^*$ is defined as

$$\delta_A(L) \stackrel{\text{def}}{=} \lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} \frac{\#\left(L \cap A^k\right)}{\#(A^k)}$$

if it exists, otherwise we write $\delta_A(L) = \bot$. The language L is called *null* if $\delta_A(L) = 0$, and dually, L is called *co-null* if $\delta_A(L) = 1$.

▶ **Example 2.** It is known that every regular language has a rational density (*cf.* [1]) and it is computable. For each word w, the language A^*wA^*, the set of all words that contain w as a factor, is of density one (co-null). This fact follows from the so-called *infinite monkey theorem*: take any word w. A random word of length n contains w as a factor with probability tending to 1 as $n \to \infty$.

We list some basic properties of the density as follows.

▶ **Lemma 3.** *Let $K, L \subseteq A^*$ with $\delta_A(K) = \alpha, \delta_A(L) = \beta$. Then we have:*
1. *$\delta_A(L \setminus K) = \beta - \alpha$ if $K \subseteq L$.*
2. *$\delta_A(K \cup L) = \alpha + \beta$ if $K \cap L = \varnothing$.*
3. *$\delta_A(uLv) = \delta_A(L) \cdot \#(A)^{-|uv|}$ for each $u, v \in A^*$.*

▶ **Lemma 4.** *Let $\eta\colon A^* \to G$ be a morphism onto a finite group. For any $g \in G$, we have $\delta_A(\eta^{-1}(g)) = \#(G)^{-1}$.*

Let L be a regular language, M_L be its syntactic monoid and $\eta : A^* \to M_L$ be its syntactic morphism. The kernel of M_L is a unique minimal ideal of M_L (it is always exists since M_L is finite). For every m not in the kernel of M_L, $\eta^{-1}(m) \cap A^*wA^* = \emptyset$ holds for some $w \in A^*$ hence $\delta_A(\eta^{-1}(m)) = 0$ by the infinite monkey theorem. For more detailed properties of δ_A, see Chapter 13 of [1].

The notion of "measurability" on formal languages is defined by a standard measure theoretic approach as follows.

▶ **Definition 5** ([14])**.** Let $\mathcal{C}_A$ be a family of languages over A. For a language $L \subseteq A^*$, we define its *$\mathcal{C}_A$-inner-density* $\underline{\mu}_{\mathcal{C}_A}(L)$ and *$\mathcal{C}_A$-outer-density* $\overline{\mu}_{\mathcal{C}_A}(L)$ over A as

$$\underline{\mu}_{\mathcal{C}_A}(L) \stackrel{\mathsf{def}}{=} \sup\{\, \delta_A(K) \mid K \subseteq L, K \in \mathcal{C}_A, \delta_A(K) \neq \bot \,\} \text{ and}$$

$$\overline{\mu}_{\mathcal{C}_A}(L) \stackrel{\mathsf{def}}{=} \inf\{\, \delta_A(K) \mid L \subseteq K, K \in \mathcal{C}_A, \delta_A(K) \neq \bot \,\}, \text{ respectively.}$$

A language L is said to be *$\mathcal{C}_A$-measurable* if $\underline{\mu}_{\mathcal{C}_A}(L) = \overline{\mu}_{\mathcal{C}_A}(L)$ holds. We say that an infinite sequence $(L_n)_n$ of languages over A *converges to L from inner (from outer,* respectively*)* if $L_n \subseteq L$ ($L_n \supseteq L$, respectively) for each n and $\lim_{n\to\infty} \delta_A(L_n) = \delta_A(L)$. A langauge L is $\mathcal{C} = A$-measurable if and only if there exist a convergent sequence of languages in $\mathcal{C}_A$ to L both from inner and outer.

The following is an example of Gcom-measurable non-regular language.

▶ **Example 6** ([14])**.** The semi-Dyck language $\mathsf{D} = \{\varepsilon, ab, aabb, abab, aaabbb, \ldots\}$ over $A = \{a, b\}$ is Gcom-measurable. For each $k \geq 2$, the language $L_k = \{w \in A^* \mid |w|_a = |w|_b \bmod k\}$ is in Gcom, because $\eta^{-1}(0) = L_k$ holds for the morphism $\eta\colon A^* \to \mathbb{Z}/k\mathbb{Z}$ where $h(a) = 1$ and $h(b) = k - 1$. Obviously, $\mathsf{D} \subseteq L_k$ holds and it follows from Lemma 4 that $\delta_A(L_k) = 1/k$ holds. Hence $\delta_A(L_k)$ tends to zero if k tends to infinity.

For a family $\mathcal{C}_A$ of languages over A, we denote by $\mathrm{Ext}(\mathcal{C}_A)$ ($\mathrm{RExt}(\mathcal{C}_A)$, respectively) the class of all $\mathcal{C}_A$-measurable languages ($\mathcal{C}_A$-measurable *regular* languages, respectively) over A.

▶ **Lemma 7** ([15])**.** *The operator* Ext *is a closure, i.e., it satisfies the following three properties for each $\mathcal{C} \subseteq \mathcal{D} \subseteq 2^{A^*}$: (extensive) $\mathcal{C} \subseteq \mathrm{Ext}(\mathcal{C})$, (monotone) $\mathrm{Ext}(\mathcal{C}) \subseteq \mathrm{Ext}(\mathcal{D})$, and (idempotent) $\mathrm{Ext}(\mathrm{Ext}(\mathcal{C})) = \mathrm{Ext}(\mathcal{C})$. Moreover, $\mathrm{Ext}(\mathcal{C}_A)$ is closed under Boolean operations and quotients if $\mathcal{C}_A$ is closed under Boolean operations and quotients.*

3 Measuring Power of Star-Free Languages

In this section, we give a proof of the first main theorem, which gives a decidable characterisation of SF-measurability for regular languages.

▶ **Theorem 8.** *The class of star-free languages and the class of generalised definite languages have the same measuring power, i.e.,* $\mathrm{Ext}(\mathsf{SF}) = \mathrm{Ext}(\mathsf{GD})$.

To prove Theorem 8, we require some lemmata, where Lemma 11 is the key of our proof.

▶ **Lemma 9** ([17])**.** *Every regular language whose density is zero or one is* GD*-measurable.*

▶ **Lemma 10.** *Let* $\mathcal{C} \subseteq \mathrm{REG}_A$ *be a class of regular languages closed under Boolean operations,* M *be a finite monoid,* $\alpha : A^* \to M$ *be a monoid morphism and* $L \subseteq A^*$ *be a langauge such that* $L = \alpha^{-1}(\alpha(L))$*. If there is a convergent sequence of languages in* $\mathcal{C}$ *to* $\alpha^{-1}(m)$ *from inner for every* $m \in M$*, then* L *is* $\mathcal{C}$*-measurable.*

Proof. For each $m \in M$, let $(L_{m,n})_n$ be a convergent sequence of languages in $\mathcal{C}$ to $\alpha^{-1}(m)$ from inner. Since $\mathcal{C}$ is closed under finite union, the sequence $(\bigcup_{m \in \alpha(L)} L_{m,n})_n$ is a convergent sequence of languages in $\mathcal{C}$ to L. Since $\mathcal{C}$ is closed under finite intersection and complementation, the sequence $(\bigcap_{m \notin \alpha(L)} L_{m,n}^{\mathrm{c}})_n$ is a convergent sequence of languages in $\mathcal{C}$ to L. ◀

▶ **Lemma 11.** *Let* L *be a regular language and* M_L *be its syntactic monoid. If every* $\mathcal{H}$*-class in the kernel of* M_L *is trivial, then* L *is* GD*-measurable.*

Proof. Let $\eta : A^* \to M$ be the syntactic morphism of L. Let K be the kernel of M_L. If $m \notin K$, then the infinite monkey theorem implies $\delta_A(L_m) = 0$ hence L_m is GD-measurable by Lemma 9. Let $m \in M$ be an element in K. In this case, we have $\mathcal{H}_m = mM \cap Mm$. By the assumption, K is $\mathcal{H}$-trivial which implies $\mathcal{H}_m = \{m\} = mM \cap Mm$. This means that

$$L_m = \eta^{-1}(mM \cap Mm) = \eta^{-1}(mM) \cap \eta^{-1}(Mm) = L_m A^* \cap A^* L_m.$$

For each $n \geq 0$, we define a generalised definite language $L_m^{\leq n}$ as

$$L_m^{\leq n} = \{w \in L_m \mid |w| \leq n\}A^* \cap A^*\{w \in L_m \mid |w| \leq n\}.$$

It is easy to see that the sequence of generalised definite languages $(L_m^{\leq n})_{n \geq 0}$ satisfies $L_m \supseteq L_m^{\leq n}$ for each n and $\lim_{n \to \infty} \delta_A(L_m^{\leq n}) = \delta_A(L_m)$, *i.e.*, $(L_m^{\leq n})_{n \geq 0}$ is a convergent sequence to L_m from inner. We showed that, for every $m \in M$, there is a convergent sequence of generalised definite languages to L_m from inner. Hence L is GD-measurable by Lemma 10 because GD is closed under Boolean operations. ◀

Proof of Theorem 8. Since we have $\mathsf{GD} \subseteq \mathsf{SF}$, it is enough to show that $\mathsf{SF} \subseteq \mathrm{Ext}(\mathsf{GD})$ which implies $\mathrm{Ext}(\mathsf{GD}) \subseteq \mathrm{Ext}(\mathsf{SF}) \subseteq \mathrm{Ext}(\mathrm{Ext}(\mathsf{GD})) = \mathrm{Ext}(\mathsf{GD})$ by Lemma 7. The syntactic monoid of any star-free language is $\mathcal{H}$-trivial, hence $\mathsf{SF} \subseteq \mathrm{Ext}(\mathsf{GD})$ holds by Lemma 11. ◀

Actually, the converse direction of Lemma 11 holds.

▶ **Theorem 12.** *Let* L *be a regular language and* M_L *be its syntactic monoid. Then* L *is* GD*-measurable if and only if every* $\mathcal{H}$*-class in the kernel of* M_L *is trivial.*

Proof. TODO. ◀

4 Measure-Theoretic Decomposition Theorem of Regular Languages

In this section, we give a proof of the second main theorem.

▶ **Theorem 13.** *Every regular language is* $\mathscr{B}(\mathsf{SF} \cup \mathsf{G}))$*-measurable, i.e.,* $\mathrm{REG} = \mathrm{REXT}(\mathscr{B}(\mathsf{SF} \cup \mathsf{G}))$.

Proof. Let $L \subseteq A^*$ be regular language, M_L be its syntactic monoid, K be the kernel of M_L and $\eta : A^* \to M_L$ be its syntactic morphism. By the same argument in the proof of Theorem 8, there is a convergent sequence of generalised definite langauges to $\eta^{-1}(m)$ from inner for each $m \notin K$. Let $m \in K$ and define a morphism $\eta_\downarrow : A^* \to \mathcal{H}_m$ as $\eta_\downarrow(a) = e_m\eta(a)e_m$ where e_m is the identity element of the group $\mathcal{H}_m = mM \cap Mm = e_mMe_m$. We now show that $\eta^{-1}(m) = \eta^{-1}(\mathcal{H}_m) \cap \eta_m^{-1}(m)$ holds. To prove $\eta^{-1}(m) \subseteq \eta_\downarrow^{-1}(m)$, let $a_1 \cdots a_n \in \eta^{-1}(m)$ and $\eta(a_i) = x_i$ for each i. ◀

▶ **Corollary 14.** *A language* L *is* REG*-measurable if and only if* L *is* $\mathscr{B}(\mathsf{SF} \cup \mathsf{G}))$*-measurable.*

5 Some Applications of Two Main Theorems

5.1 Axiomatic characterisation of the density function δ_A

5.2 Probabilistic independence of star-free and group languages

5.3 A simple proof of the existence of non-primitive words in a regular language with positive density

6 Conclusion and Open Problems

... While the AT-measurability is strictly weaker than PT-measurability, it is interesting that the computational complexity of the AT-measurability is much higher than PT-measurability: deciding the AT-measurability of a given regular language L (represented by a deterministic automaton) is *PSPACE-complete* but deciding the PT-measurability of L can be done *in linear time* with respect to the number of states [18]. ...

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